Bihar Engineering University, Patna

B.Tech. 3rd Semester Examination, 2023

Course: B.Tech. **Time: 03 Hours**
Code: 101304 **Subject: Engineering Mechanics** **Full Marks: 70**

Instructions:-
(i) *The marks are indicated in the right-hand margin.*
(ii) *There are **NINE** questions in this paper.*
(iii) *Attempt **FIVE** questions in all.*
(iv) *Question No. **1** is compulsory.*

Q.1 ***Choose the correct answer of the following (any seven question only):*** **[2 x 7 = 14]**

(a) Two forces P and Q are acting at an angle α. The resultant force R acts at an angle of θ with force P, then the value of θ will be
(i) $tan^{-1}\left(\frac{Q\cos\alpha}{P+Q\cos\alpha}\right)$ (ii) $tan^{-1}\left(\frac{Q\sin\alpha}{P+Q\cos\alpha}\right)$
(iii) $tan^{-1}\left(\frac{Q\cos\alpha}{P+Q\sin\alpha}\right)$ (iv) $tan^{-1}\left(\frac{Q\sin\alpha}{P+Q\sin\alpha}\right)$

(b) Three forces P, Q and R are acting concurrently. The included angles are α, β and γ. According to Lami's theorem
(i) $\frac{P}{\sin\alpha}=\frac{Q}{\sin\beta}=\frac{R}{\sin\gamma}$ (ii) $\frac{P}{\cos\alpha}=\frac{Q}{\cos\beta}=\frac{R}{\cos\gamma}$
(iii) $\frac{P}{\tan\alpha}=\frac{Q}{\tan\beta}=\frac{R}{\tan\gamma}$ (iv) $\frac{\cos\alpha}{P}=\frac{\cos\beta}{Q}=\frac{\cos\gamma}{R}$

(c) If the algebraic sum of all the forces acting on a body is zero, then the body may be in equilibrium provided the forces are
(i) parallel (ii) like parallel (iii) unlike parallel (iv) concurrent

(d) Varignon's theorem is related to
(i) Principle of moments (ii) Principle of momentum
(iii) Principle of force (iv) Principle of inertia

(e) The coefficient of friction is the ratio of
(i) the normal reaction to the limiting force of friction
(ii) The weight of the body to limiting force of friction
(iii) the limiting force of friction to the normal reaction
(iv) the weight of the body to the normal reaction

(f) Moment of inertia of a body does not depend on
(i) Axis of rotation of the body
(ii) Mass of the body
(iii) Angular velocity of the body
(iv) distribution of mass in the body

(g) The moment of inertia of a rectangular section of base (b), height (h) about CG will be:
(i) $\frac{hb^3}{12}$ (ii) $\frac{bh^3}{12}$ (iii) $\frac{hb^3}{3}$ (iv) $\frac{bh^3}{3}$

(h) The condition for a truss to be perfect is where m = number of members and J = number of joints
(i) m = 2j - 3 (ii) m > 2j – 3 (iii) m < 2j - 3 (iv) m ≥ 2j - 3

(i) The linear velocity of a body rotating at ω rad/s along a circular path of radius r is
(i) ω. r (ii) ω/r (iii) ω. 2 r (iv) ω / r

(j) The units of energy in SI units
(i) Joule (ii) Watt (iii) Joule/sec. (iv) Watt/sec

Q.2 (a) State and prove parallelogram law of forces. **[4]**

(b) A horizontal line PQRS is 12 m long, where PQ=QR=RS=4 m, the forces are acting at P, Q, R and S as shown in figure 1. Find the magnitude, direction and position of the resultant force. **[10]**

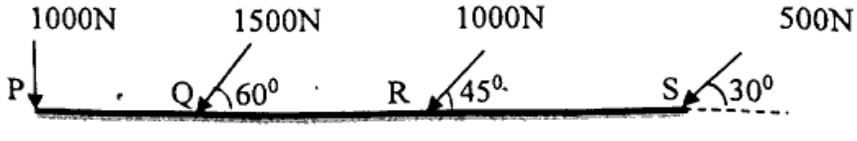


Fig. (1)

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Q.3 (a) What is moment of a couple? Why it is called free Vector? [4]

(b) A system of force and couple acting on a simply supported beam as shown in figure 2. Find the reaction force acting at its support A and E. [10]

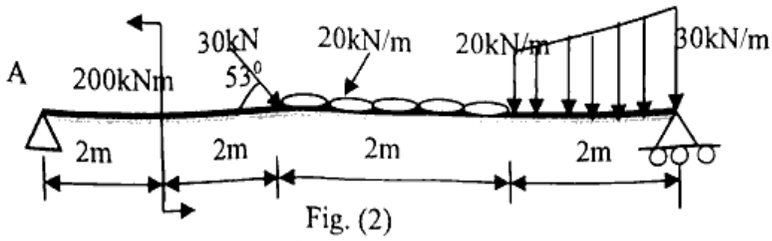


Fig. (2)

Q.4 (a) Explain the conditions of Equilibrium. [4]

(b) Two cylinders P and Q rest in a channel as shown in figure 3. The cylinder P has diameter of 100 mm and weighs 200N, where as the cylinder Q has diameter of 180 mm and weighs 500N.Determine the pressure at all the contact points. [10]

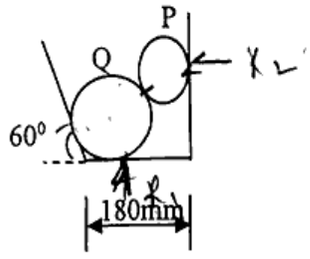


Fig. (3)

Q.5 (a) State the laws of friction. Explain the term angle of friction. [6]

(b) Two blocks A and B of weights 1 kN and 2kN respectively are in equilibrium position as shown in figure. 4. [8]

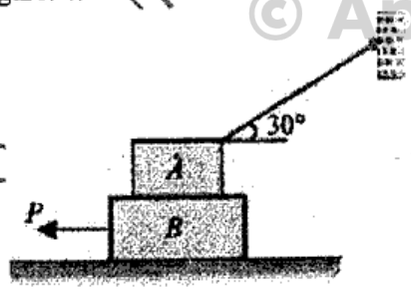


Fig. (4)

If the coefficient of friction between the two blocks as well as the block B and the floor is 0.3. find the force P required to move the block B.

Q.6 (a) Explain the principle of Virtual work. [4]

(b) [10]

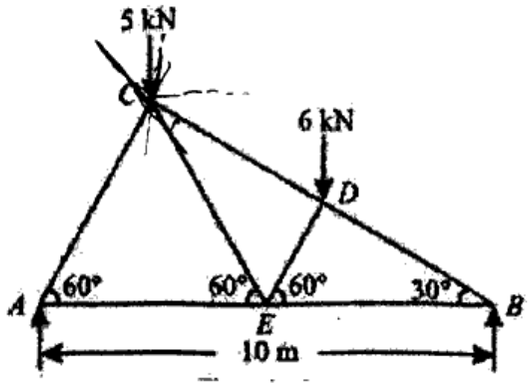


Fig. (5)

Find the reaction and forces in the members of the truss shown in figure 5.

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Q.7 (a) The equation of motion of a particle moving in a straight line is given by $S= 18t+3t^2-2t^3$, where (s) is in metres and (t) in seconds. Find (1) Velocity and acceleration at start, (2) time, when the particle reaches its maximum velocity, and (3) maximum velocity of the particle. **[7]**

(b) The motion of a particle is given by $a= t^3-3t^2+5$, where (a) is the acceleration in m/s^2 and (t) is the time in seconds. The velocity of the particle at t=1 second is 6.25m/sec and the displacement is 8.8 metres. Calculate the displacement and velocity at t=2 seconds. **[7]**

Q.8 (a) The equation for angular displacement of a particle, moving in a circular path (radius 200m) is given by $\theta= 18t+3t^2-2t^3$, where θ is the angular displacement at the end of t sec. Find (i) angular velocity and acceleration at the start, (ii) time, when the particle reaches its maximum angular velocity, and (iii) maximum angular velocity of the particle. **[6]**

(b) An I- section shown in figure 6. Find the moment of inertia of the section about the horizontal axis passing through the centre of gravity of the section. **[8]**

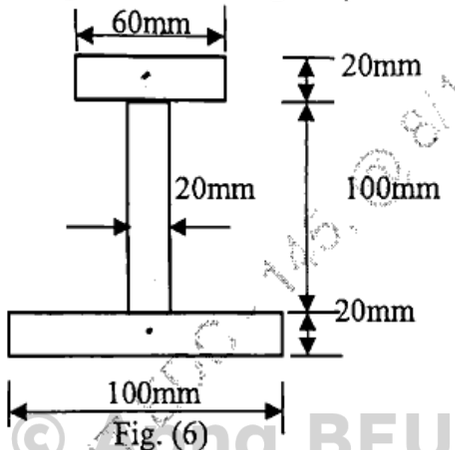
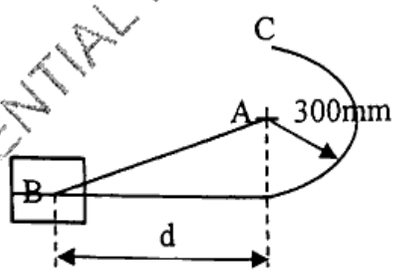


Fig. (6)

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Q.9 (a) A collar B having mass 100 g moves along a frictionless curved rod in a vertical plane. A light rubber band connects B to fixed point A. The rubber band is 250 mm in length when unstretched. A force of 30 N is required to extend the band 50 mm. The collar is released from rest. what distance must d be so that the downward normal force on the rod at C is 20 N? **[10]**



(b) Define Linear momentum and linear Impulse of a particle of mass m acted upon by a force F. **[4]**

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